DG INTERN HUB

SOC ANALYST INTERNSHIP

WEEK 2 —NOTES

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DNS Resolution

Learning Objective

To understand how DNS converts a domain name into an IP address and how a SOC analyst can use DNS activity during an investigation.

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1. What is DNS?

DNS stands for **Domain Name System**.

It is used to convert a human-readable domain name into an IP address.

For example:

www.example.com → 93.184.216.34

Instead of remembering an IP address for every website, we can use a domain name.

DNS works somewhat like a phonebook. We search for a domain name, and DNS helps us find the IP address connected to it.

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2. How DNS Resolution Works

When I type a website address into my browser, several steps can happen before the website opens.

Step 1 — I enter the domain name

For example:

www.example.com

The browser needs to know the IP address of the web server before it can connect to it.

Step 2 — Browser checks its cache

The browser may already have the IP address stored from a previous DNS lookup.

If the answer is available in the browser cache, it may not need to perform a new DNS query.

Step 3 — Operating system checks its DNS cache

If the browser does not have the answer, the operating system may check its own DNS cache.

If the IP address is already stored there, the system can use that result.

Step 4 — Query is sent to the DNS resolver

If the IP address is not found locally, the computer sends a DNS query to its configured DNS resolver.

This is usually provided by:

- Internet Service Provider
- Organization/network
- Public DNS service

The resolver is responsible for finding the answer.

Step 5 — Resolver checks its cache

The DNS resolver may already know the answer.

If the answer is cached and has not expired, it can return the IP address directly.

If it does not have the answer, it needs to find it.

Step 6 — Resolver contacts the Root DNS Server

The resolver sends a query to a **Root DNS server**.

The root server does not normally provide the final IP address.

Instead, it tells the resolver where to find the appropriate **Top-Level Domain (TLD) server**.

For example, for:

example.com

the relevant TLD is:

.com

Step 7 — TLD DNS Server

The resolver then contacts the `.com` TLD server.

The TLD server knows which **authoritative nameserver** is responsible for the domain.

It gives the resolver information about that authoritative nameserver.

Step 8 — Authoritative DNS Server

The resolver then contacts the authoritative nameserver for the domain.

The authoritative nameserver contains the actual DNS records for the domain.

It can return the IP address associated with the domain.

For example:

www.example.com → 93.184.216.34

Step 9 — Resolver sends the answer back

The DNS resolver sends the IP address back to the user's computer.

The result may also be cached for a period of time based on the DNS record's TTL.

Step 10 — Browser connects to the server

Now the browser knows the IP address.

It can establish a connection to the server and request the website.

For HTTPS, the browser will normally connect to the server using **TCP port 443**.

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3. DNS Resolution Flow

The overall process can be represented like this:

Browser
↓
Browser Cache
↓
OS DNS Cache
↓
DNS Resolver
↓
Root DNS Server

↓

TLD DNS Server

↓

Authoritative DNS Server

↓

IP Address

↓

Browser connects to Web Server

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4. Example

Suppose I type:

www.example.com

The system needs to find the IP address for that domain.

The resolver may go through:

Root Server

↓

.com TLD Server

↓

Authoritative Nameserver

↓

IP Address

After receiving the IP address, my browser can connect to the web server.

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5. Why DNS is Important for a SOC Analyst

DNS information can be useful during security investigations.

A SOC analyst can investigate:

- Which domain was requested?
- Which device made the request?
- Which user was using the device?
- What was the destination IP?
- How often was the domain requested?
- Is the domain suspicious?
- Was the domain recently created?
- Are multiple systems requesting the same unusual domain?

For example, if a system suddenly starts making repeated DNS requests to a suspicious domain, I would investigate that activity further.

DNS can also be involved in malware communication and other suspicious activity, so DNS logs can be useful during threat hunting.

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6. Important DNS Terms

Term	Meaning
DNS Resolver	Server that performs DNS lookups for clients
Root Server	Directs queries toward the correct TLD servers
TLD Server	Handles domains such as .com, .org, .net, etc.
Authoritative Nameserver	Provides the actual DNS records for a domain

DNS Cache	*Stores previous DNS results for a period of time*
TTL	*Tells how long a DNS record can be cached*

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7. My Understanding

I understood DNS as the process that helps my computer find the IP address of a website from its domain name.

The important flow for me to remember is:

Browser → Cache → DNS Resolver → Root → TLD → Authoritative Nameserver → IP Address

From a SOC point of view, DNS is also useful because the domains a system contacts can give clues about what that system is doing.

So while investigating network activity, I should not only look at IP addresses. I should also check the DNS queries and domains involved.